Experiment 6

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Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyse how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Software Requirements

- **Database Management System:**
  - o Oracle Database Express Edition (Oracle XE)
  - o PostgreSQL Database
- **Database Administration Tool / Client Tool:**
  - o Oracle SQL Developer (for Oracle XE)
  - o pgAdmin (for PostgreSQL)

Objectives

- To implement and analyse the use of implicit cursors, explicit cursors, and cursor attributes for processing multiple rows from a database table and applying business logic effectively.

Problem Statement

• In real-world applications, database queries often return multiple rows that need to be processed individually to apply specific business rules.

Practical/Experiment Steps

• **Schema Provisioning:** Created the **Students** table with attributes for student ID, name, course, and marks to simulate an academic database environment.

• **Implicit Cursor Analysis:** Implemented an anonymous PL/SQL block to perform a single-row UPDATE operation (increasing marks), utilizing the internal SQL cursor to monitor DML success.

• **Explicit Cursor Design:** Defined a named cursor (c_bca) to fetch records of students enrolled in the 'BCA' course for row-by-row processing.

• **Attribute Integration:** Used cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN to control execution flow.

• **Business Logic Application:** Applied logic to increase marks and display student-specific details, demonstrating how rules are applied to individual records.

Procedure

• Logged into the database and enabled SERVEROUTPUT to display results.

• Executed the CREATE TABLE and INSERT statements to create and populate the Students table.

• Ran a PL/SQL block using an implicit cursor to update marks of a specific student and verified using SQL%ROWCOUNT.
• Declared an explicit cursor to retrieve students belonging to the 'BCA' course.
• Opened the cursor and verified using %ISOPEN.
• Used a LOOP to fetch records into variables (v_name, v_course).
• Applied %NOTFOUND to terminate the loop after all rows were processed.
• Used %ROWCOUNT to display the number of records fetched.
• Closed the cursor and verified closure using NOT c_bca%ISOPEN.

Input/Output Analysis

SQL Input Queries

```
CREATE TABLE Students(
   student_id NUMBER PRIMARY KEY,
   name VARCHAR(20),
   course VARCHAR(20),
   marks NUMBER
);
```

Output

```
SQL> CREATE TABLE Students(
         student_id NUMBER PRIMARY KEY,
         name VARCHAR(20),
         course VARCHAR(20),...
Show more...
```

```
Table STUDENTS created.

Elapsed: 00:00:00.019
```

SQL Input Queries

INSERT INTO Students VALUES (1, 'Rahul', 'BCA', 70);
INSERT INTO Students VALUES (2, 'Anita', 'BBA', 65);
INSERT INTO Students VALUES (3, 'Karan', 'BCA', 80);
INSERT INTO Students VALUES (4, 'Neha', 'BBA', 75);
COMMIT;

Output

```
SQL> INSERT INTO Students VALUES (2, 'Anita', 'BBA', 65)

1 row inserted.
Elapsed: 00:00:00.002
```

```
SQL> INSERT INTO Students VALUES (3, 'Karan', 'BCA', 80)

1 row inserted.
Elapsed: 00:00:00.001
```

```
SQL> INSERT INTO Students VALUES (4, 'Neha', 'BBA', 75)

1 row inserted.
Elapsed: 00:00:00.001
```

SQL Input Queries

```
SET SERVEROUTPUT ON;

DECLARE
   v_target_id NUMBER := 1;

BEGIN
   UPDATE Students
   SET marks = marks * 1.05
   WHERE student_id = v_target_id;

   IF SQL%FOUND THEN
      DBMS_OUTPUT.PUT_LINE('SQL%FOUND is TRUE');
      DBMS_OUTPUT.PUT_LINE('Rows updated: ' || SQL%ROWCOUNT);
   ELSE
      DBMS_OUTPUT.PUT_LINE('No row updated');
   END IF;
```

END;

Output

```
SQL> DECLARE
         v_target_id NUMBER := 1;

     BEGIN ...
Show more...


SQL%FOUND is TRUE
Rows updated: 1


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.015
```

SQL Input Queries

```
DECLARE
   CURSOR c_bca IS
      SELECT name, course FROM Students WHERE course
= 'BCA';

   v_name Students.name%TYPE;
   v_course Students.course%TYPE;
```

```
BEGIN

   OPEN c_bca;

   IF c_bca%ISOPEN THEN
      DBMS_OUTPUT.PUT_LINE('Cursor is OPEN');
   END IF;

   LOOP
      FETCH c_bca INTO v_name, v_course;

      IF c_bca%NOTFOUND THEN
         DBMS_OUTPUT.PUT_LINE('No more records');
         EXIT;
      END IF;

      DBMS_OUTPUT.PUT_LINE('Student #' ||
c_bca%ROWCOUNT || ': '
      || v_name || ' | ' || v_course);

   END LOOP;

   CLOSE c_bca;

   IF NOT c_bca%ISOPEN THEN
      DBMS_OUTPUT.PUT_LINE('Cursor CLOSED
successfully');
   END IF;

END;
```

Output

```
SQL> DECLARE
         CURSOR c_bca IS
             SELECT name, course FROM Students WHERE course = 'BCA';
     ...
Show more...


Cursor is OPEN
Student #1: Rahul | BCA
Student #2: Karan | BCA
No more records
Cursor CLOSED successfully


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.114
```

Learning Outcomes

• Gained understanding of implicit cursors used in DML operations.
• Learned how explicit cursors process multiple rows sequentially.
• Understood the use of %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN in controlling program flow.
• Learned the importance of OPEN, FETCH, and CLOSE operations for cursor management.
• Developed the ability to apply business logic to individual records in PL/SQL.